Paper Code:

Measurement-Oriented Microscopy Reconstruction for Cross-Magnification Micro-Displacement Inspection

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Introduction

Optical inspection benefits from reconstruction when clearer boundaries produce more stable position measurements. Changing magnification changes object-space sampling and the pixel phase of the same motion. Single-step displacement dispersion therefore provides a direct test of measurement utility.

MOSR reconstructs each monochrome image on its original pixel grid. One fixed set of learned weights serves 20× and 50× objectives. Raw, fixed Wiener and MOSR use the same image pairs and pillar identities.

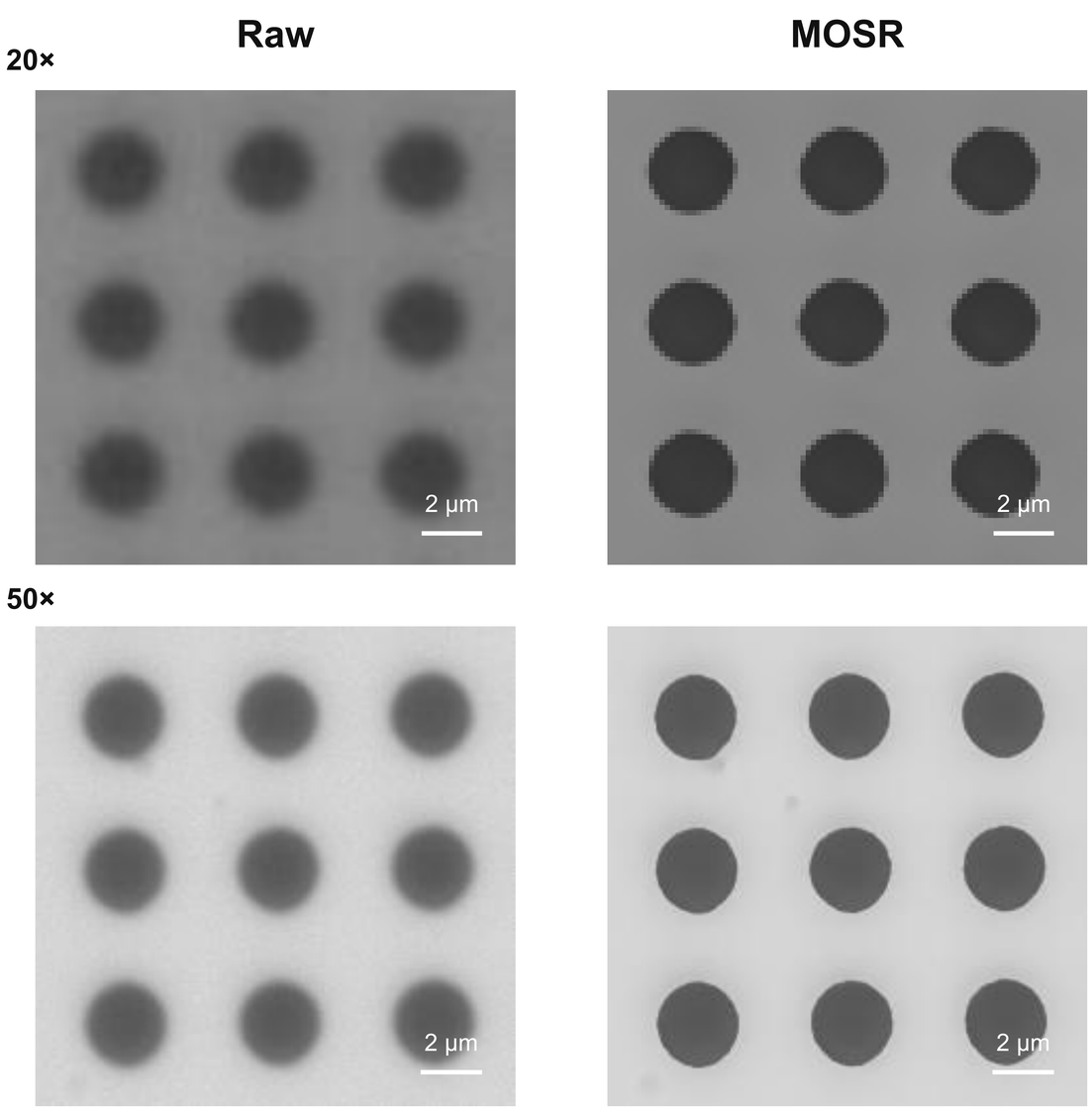


Fig. 1. Native-grid Raw and MOSR pillar fields at 20× and 50×. The common display range preserves direct visual comparison. Scale bars: 2 μm.

Methods

Reconstruction. MOSR combines U-Net context [1], anti-aliasing [2] and bounded PSF-guided correction. Optical inputs comprise 25 measured kernels at 20× and one calibrated kernel at 50×.

Localization. Coarse contours [3] initialize geometric least-squares circle (LSC) centers [4]. Each pillar uses 720 uniform radial directions, 0.05-pixel profile sampling and three center updates.

Displacement. Each pair yields nine horizontal pillar-center shifts. Their median, converted to object space and taken in absolute value, defines one single-step displacement. All three image methods use the same procedure.

TABLE I. FIXED ACQUISITION CONDITIONS

Objective	Step (μm)	Pairs	μm/pixel
50×	0.5	30	0.0690
50×	1.0	30	0.0690
20×	0.5	30	0.1725
20×	1.0	30	0.1725

Within each condition, Step SD is the sample standard deviation of its 30 single-step displacements (denominator 29). The reported mean is the arithmetic mean of the four condition-level SDs.

Results

Each condition contributes one Step SD from 30 measured single steps. MOSR yields the lowest Step SD in all four conditions, directly linking reconstruction to reduced step-to-step dispersion.

19.5%

8.84 nm Raw
7.11 nm MOSR

Reduction in the mean Step SD

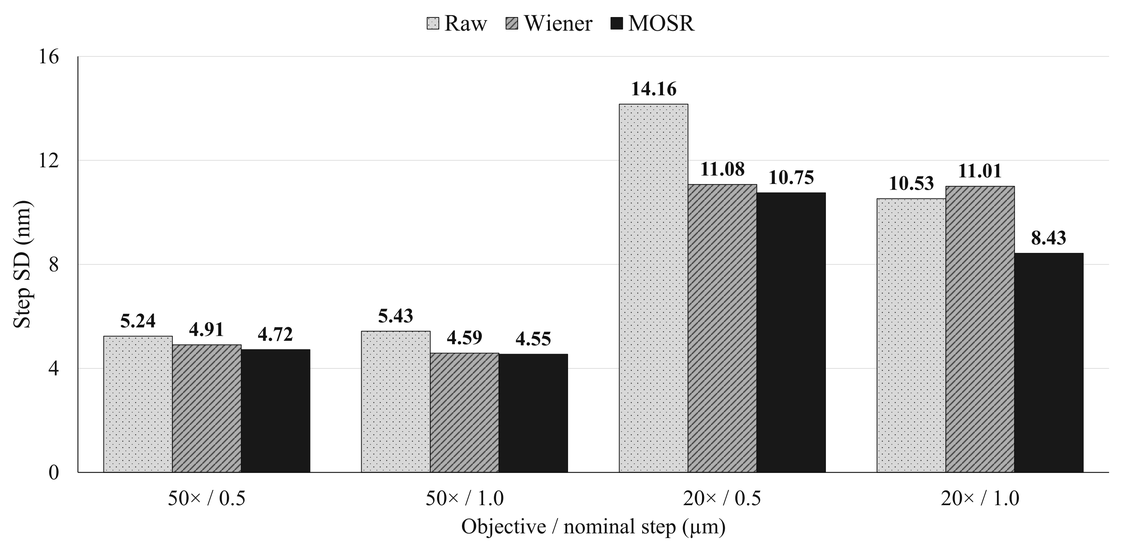


Fig. 2. Condition-level standard deviation of 30 single-step displacements. All methods use the same locked pairs and circle-center reader.

The reduction relative to Raw spans 9.8–24.1% across the four conditions. The largest reduction occurs at 20× with 0.5 μm steps, where Step SD decreases from 14.16 to 10.75 nm. MOSR also improves both 50× conditions.

TABLE II. STEP SD AND RELATIVE REDUCTION

Condition	Raw (nm)	Wiener (nm)	MOSR (nm)	Reduction vs Raw (%)
50× / 0.5	5.24	4.91	**4.72**	9.8
50× / 1.0	5.43	4.59	**4.55**	16.3
20× / 0.5	14.16	11.08	**10.75**	24.1
20× / 1.0	10.53	11.01	**8.43**	19.9
Mean	**8.84**	**7.90**	**7.11**	**19.5**

Mean row: arithmetic mean of the four SDs. Reduction = (Raw − MOSR) / Raw.

Conclusions

MOSR lowers single-step dispersion in every tested condition, with reductions of 9.8–24.1% relative to Raw. The mean condition-level Step SD decreases by 19.5%.

One set of reconstruction weights supports both objectives. Clearer pillar boundaries and lower displacement dispersion establish MOSR as a measurement-oriented reconstruction approach for cross-magnification optical inspection.

Bibliography

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